

Case Study: Customer Segmentation Using a Feedforward Neural Network

1. Problem Description and Rationale

This case study demonstrates the design and application of a custom-built feedforward neural network library to solve a real-world binary classification problem: customer segmentation. The objective is to classify customers into meaningful segments based on demographic and professional attributes.

Customer segmentation is widely used in marketing, recommendation systems, and business analytics. This problem was selected because it involves mixed data types, requires preprocessing, and provides a realistic environment to validate forward propagation, backpropagation, and loss optimization.

2. Dataset Description and Preprocessing

The dataset consists of 1000 customer records stored in a CSV file. Each record includes age, gender, occupation, region, and a binary target label.

Preprocessing steps included one-hot encoding for categorical variables, min-max normalization for numerical features, handling missing or invalid values gracefully, and splitting the dataset into 75% training and 25% testing subsets.

3. Neural Network Architecture

The final architecture consists of an input layer, two hidden layers (16 and 8 neurons respectively) using (ReLU/Sigmoid/Tanh/Linear) activation, and a single-neuron output layer. (He/Xavier/RandomUniform) initialization was used for hidden layers.

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4. Training and Evaluation Results

The model was trained using gradient descent optimization with a learning rate of 0.01, batch size of 32, and 1000 epochs. Cross-entropy loss was used.

Training loss decreased steadily across epochs, and the final test accuracy ranged between 80% and 85%.

5. Library Usage Explanation

The neural network library was used to load and preprocess data, define a customizable architecture, compile the model with a selected loss function and optimizer, train the model, and evaluate predictions. The library design ensured separation of concerns and avoided problem-specific logic.

6. Supporting Results

Training loss trends, prediction outputs, and evaluation summaries were printed during execution. These outputs validate correct learning behavior and successful use of the neural network framework.